Physical Therapy in Sport 16 (2015) 16–21

Contents lists available at ScienceDirect

Physical Therapy in Sport

journal homepage: www.elsevier.com/ptsp

Original research

# Effect of lace-up ankle braces on electromyography measures during walking in adults with chronic ankle instability

Greg Barlow [1], Luke Donovan*, Joseph M. Hart [2], Jay Hertel [3]

*University of Virginia, 210 Emmet St. South, Charlottesville, VA 22904, USA*

ARTICLE INFO

*Article history:*
Received 9 November 2013
Received in revised form
29 January 2014
Accepted 11 February 2014

*Keywords:*
Ankle sprain
Muscle activation
Neuromuscular control

ABSTRACT

*Background:* Lace-up ankle braces reduce the incidence of ankle sprains and have been hypothesized to do so through both mechanical and neuromuscular mechanisms.
*Objective:* To determine the effect of lace-up ankle braces on surface electromyography (sEMG) measures during walking in adults with chronic ankle instability (CAI).
*Design:* Randomized crossover.
*Setting:* Laboratory.
*Participants:* Fifteen adults with CAI.
*Main outcome measures:* Surface EMG activity was recorded from the anterior tibialis, peroneus longus, lateral gastrocnemius, rectus femoris, biceps femoris and gluteus medius during treadmill walking with and without lace-up ankle braces. The dependent variables were sEMG amplitude 100 ms pre- and 200 ms post-initial contact, time of activation relative to initial contact, and percent of activation across the stride cycle.
*Results:* When compared to no brace, ankle bracing resulted in lower pre-contact amplitude of the peroneus longus ($p = 0.02$). The anterior tibialis, peroneus longus, rectus femoris, and gluteus medius were activated later relative to initial contact ($p < 0.03$). The peroneus longus and rectus femoris were activated for a shorter percentage of the stride cycle ($p < 0.05$).
*Conclusion:* Braces cause a change in neuromuscular activity during walking. Clinicians should be aware of these changes when prescribing braces, as it may relate to the mechanism in which braces decrease sprains.

© 2014 Elsevier Ltd. All rights reserved.

## 1. Introduction

Injuries to the ankle are one of the most common musculoskeletal injuries in athletics (Hootman, Dick, & Agel, 2007; Waterman, Owens, Davey, Zacchilli, & Belmont, 2010). People who sprain their ankle will remain about 5 times more likely to re-injure their ankle (McKay, Goldie, Payne, & Oakes, 2001). It is common for individuals who sprain their ankle to have prolonged symptoms and perceived instability that lasts greater than a year (van Rijn, van Os, Bernsen, Luijsterburg, Koes, & Bierma-Zeinstra, 2008). This condition is commonly known as chronic ankle instability (CAI) (Delahunt, Coughlan, Caulfield, Nightingale, Lin, & Hiller, 2010; Hertel, 2002). These symptoms can lead to decreased athletic performance, prolonged removal from participation and a lowered quality of life (van Rijn et al., 2008).

It can be a challenge for clinicians to improve function in people with CAI because of the complexity of the characteristics. Past studies have shown individuals with CAI to have deficits in proprioception (Munn, Sullivan, & Schneiders, 2010), postural control (Docherty, Valovich McLeod, & Shultz, 2006; Evans, Hertel, & Sebastianelli, 2004), lower extremity muscle activity (Delahunt, Monaghan, & Caulfield, 2006a; Feger, Donovan, Hart, & Hertel, 2014) and have been found to have alterations in gait (Chinn, Dicharry, & Hertel, 2013; Delahunt et al., 2006a) and landing activities (Delahunt, Monaghan, & Caulfield, 2006b). The etiology of CAI is clearly a multifactorial issue (Freeman, Dean, & Hanham, 1965; Hertel, 2008; Hiller, Kilbreath, & Refshauge, 2011).

Prophylactic ankle bracing has been used to prevent initial and recurrent injuries to the lateral ankle ligaments and is often used by individuals with CAI to enhance ankle stability. Recent studies have shown the use ankle braces caused a 70% reduction in acute ankle

* Corresponding author. Tel.: +1 434 924 6184; fax: +1 434 924 1389.
*E-mail addresses:* gbarlow39@gmail.com (G. Barlow), ltd2r@virginia.edu (L. Donovan), jmh3zf@virginia.edu (J.M. Hart), jnh7g@virginia.edu (J. Hertel).
[1] Tel.: +1 434 924 6184; fax: +1 434 924 1389.
[2] Tel.: +1 434 924 6187; fax: +1 434 924 1389.
[3] Tel.: +1 434 243 8673; fax: +1 434 924 1389.

http://dx.doi.org/10.1016/j.ptsp.2014.02.002
1466-853X/© 2014 Elsevier Ltd. All rights reserved.

injuries among those with previous ankle injury and a 57% reduction among those with no history of ankle injury in high school athletes (McGuine, Brooks, & Hetzel, 2011; McGuine, Hetzel, Wilson, & Brooks, 2012). An interesting finding from these studies (McGuine et al., 2011, 2012) is that the severity of the ankle sprains that occurred did not change with the application of a brace.

Previous literature demonstrating that ankle bracing restricts range of motion measurements (DiStefano, Padua, Brown, & Guskiewicz, 2008; Sefton, Hicks-Little, Koceja, & Cordova, 2007) have been performed in a laboratory and may not directly relate to the ability for an ankle brace to decrease range of motion when a lateral ankle injury mechanism occurs during high velocity functional activities. Neuromuscular adaptations also may have a contributing role in the ability to prevent ankle sprains. Studies utilizing trap door mechanisms (Cordova & Ingersoll, 2003; M. L. Cordova, Cardona, Ingersoll, & Sandrey, 2000; Shima, Maeda, & Hirohashi, 2005) or those measuring H-reflexes (Nishikawa & Grabiner, 1999; Sefton et al., 2007) have found increased peroneal reflex activity and motonueron pool excitability with ankle brace application. Increased cutaneous and joint mechanoreceptor stimulation from braces is thought to improve proprioception (Feuerbach, Grabiner, Koh, & Weiker, 1994). Gait alterations such as increased maximum pronation and pronation velocity (Nishikawa, Kurosaka, Yoshiya, Lundin, & Grabiner, 2002) and decreased plantarflexion at initial contact following a jump task (DiStefano et al., 2008) have also been associated with the use of ankle braces. These loading and biomechanical changes may result in a decreased chance of the foot being in a position more prone to ankle sprain.

The use of surface electromyography (sEMG) allows for non-invasive approximation of muscle activity. Previous literature utilizing sEMG related to ankle dynamics has primarily been focused on peroneus longus activation during an unanticipated perturbation (Cordova, Armstrong, Rankin, & Yeasting, 1998). While this data adds important baseline information, the translation of these results to more dynamic tasks is uncertain. A recent study on individuals with and without CAI found altered lower extremity muscle activation during walking (Feger et al., 2014). Specifically, CAI participants were found to activate lower extremity muscles earlier in the stride cycle and, in regards to the peroneus longus, for a longer percentage of the stride cycle compared to healthy controls (Feger et al., 2014). The ability for ankle braces to affect these alterations, however, is unknown. Therefore, the purpose of this study was to determine what effect lace-up ankle braces had on lower extremity sEMG amplitude and timing during treadmill walking in patients with CAI.

## 2. Methods

### 2.1. Design

A controlled laboratory study using a randomized crossover design was performed. The independent variable was condition, brace vs. no brace. Dependent variables were sEMG amplitude before and after initial contact, onset timing relative to initial contact, and percent activation time across the entire stride cycle of the (anterior tibialis (AT), peroneus longus (PL), lateral head of the gastrocnemius (LG), rectus femoris (RF), biceps femoris (BF), and gluteus medius (GM)). The randomization of condition order (brace vs. no brace) was generated by an investigation team member not involved in data collection.

### 2.2. Participants

Fifteen recreationally active young adults with CAI participated (Table 1). We defined CAI as anybody with a history of a lateral ankle sprain where the initial sprain occurred greater than one year prior and currently has self-reported functional deficits and symptoms of "giving way" or instability. Inclusion criteria included: aged 18–40, recreationally active (20 min or greater of moderate physical activity level at least 3 days per week), initial ankle sprain greater than 1 year prior to starting the study and diminished self-reported function quantified by a Foot and Ankle Ability Measure (FAAM) sport subscale score of less than 85% (Carcia, Martin, & Drouin, 2008). Participants were excluded if they had an ankle sprain within 6 weeks of the study, a history of fracture or surgical intervention to the ankle complex, current self-reported symptoms of other lower extremity injury, or any neurological pathology that could alter our main outcome measures. Measures were obtained unilaterally. In participants with bilateral CAI, data was collected from the participants' more symptomatic limb. The University Institutional Review Board approved this study and all participants provided informed consent prior to participation.

**Table 1**
Participant demographics.

| | CAI mean (standard deviation) |
|---|---|
| Age (years) | 23.0 ± 4.2 |
| Sex | Male: 5, female: 10 |
| Height (cm) | 173.3 ± 10.8 |
| Weight (kg) | 72.4 ± 14.0 |
| FAAM ADL (%) | 87.2 ± 7.1 |
| FAAM sport (%) | 68.5 ± 5.7 |
| Godin leisure-time exercise score | 94.3 ± 46.6 |
| # of Previous ankle sprains | 4.5 ± 3.2 |
| Time since last ankle sprain (months) | 15.2 ± 9.3 |

cm – Centimeter.
kg – Kilogram.

### 2.3. Instrumentation

Disposable pre-gelled Ag/AgCl electrodes were used to collect sEMG signals. Signals were amplified with a high-gain, differential-input biopotential amplifier with a gain of 1000 and digitized with a 16-bit data acquisition system (MP 150, Biopac Systems, Inc., Goleta, CA) at 2000 Hz with a common-mode rejection ratio of 110 dB, an input impedance of 1.0 MΩ, and a noise voltage of 0.2 mV. Acqknowledge software (v.4.0, Cambridge, England) was used for data collection and processing of sEMG signals. The sEMG data was collected using real-time processing with a 10–500 Hz band pass filter and a 10 sample moving average root mean square algorithm. A foot switch (BIOPAC Systems Inc., Santa Barbara, CA) was used to identify initial contact during walking. Subjects performed walking trials on a treadmill (Biodex, Shirley, NY). The lace-up brace used during testing was the McDavid Ultralight 195 (McDavid Inc., Woodridge, IL). This brace was chosen because it was used in a recent large scale clinical trial that demonstrated injury prevention effects (McGuine et al., 2011). All participants were fitted with the same model of athletic shoe (X755WB New Balance Athletic Shoe, Inc., Boston, MA) for both the braced and no brace conditions. Data was analyzed using Statistical Package for Social Sciences (SPSS) Version 20.0 (SPSS, Inc, Chicago, IL).

### 2.4. Procedures

Data collection measures, with the exception of ankle bracing, were identical to those previously described by Feger et al. (2014).

#### 2.4.1. Day 1

Participants completed the initial screening process including Godin Leisure-Time Exercise Questionnaire, FAAM activities of daily

living and sport subscales. Their information was reviewed and, if deemed eligible for the study, they were allowed a brief familiarization period with the ankle braces. A return appointment was then scheduled for between two and seven days later.

#### 2.4.2. Day 2

Participants reported back to the lab. The areas for sEMG pad application were determined using a sEMG placement map (Konrad, 2005). These areas were shaved, debrided, and cleansed using isopropyl alcohol. Two electrodes with an interelectrode distance of 2 cm were placed on the skin in the recognized position for each muscle parallel to muscle fiber orientation and a ground electrode was placed on the ipsilateral tibial shaft.

Prior to data collection participants performed a 5 min warm-up walking trial. Maximum isometric voluntary contraction (MVIC) measurements and quiet standing data during 10 s of bipedal stance were collected in order to generate baseline sEMG activity and normalize the data between participants (Lehman & McGill, 1999). MVICs of each muscle were collected using manual resistance with each contraction being held for 5 s. The sEMG data were also visually inspected during this time to assure appropriate electrode placement and evaluate for crosstalk (Konrad, 2005). A foot switch with negative and positive amplitudes was placed on the shoe. The negative amplitude was placed directly under the center of the heel of the shoe and the positive amplitude was placed on the shoe under the 1st metatarsal head to signal heel strike and toe off respectively.

Participants then walked on a treadmill at 0° incline at 4.83 km/h until they verbally indicated that they had reached a steady gait pattern (Chinn et al., 2013). Once this occurred, 30 s of sEMG data were collected. Following the conclusion of data collection in the first condition, the participant rested for 5 min as they changed bracing conditions. Each participant performed the walking task while braced and not braced bilaterally. Identical procedures were followed during the second condition.

### 2.5. Data processing

#### 2.5.1. Quiet standing

A 500 ms time epoch during quiet standing was selected for analysis. The root mean square (RMS) amplitude mean and standard deviation were calculated and used for normalization (Lehman & McGill, 1999).

#### 2.5.2. Pre-initial contact sEMG amplitude

A 100 ms time epoch immediately prior to initial contact was analyzed. These data were used to calculate the area under the RMS curve quantifying muscle activation amplitude pre-contact. The area under the RMS curve was normalized by dividing it by the mean RMS value of each corresponding muscle during quiet standing. A total of nine strides were processed (three consecutive steps were taken during the first, middle, and last 10 s of the treadmill walking). The average amplitude for the 9 trials was calculated and subjected to statistical analysis.

#### 2.5.3. Post-initial contact sEMG amplitude

A 200 ms time epoch immediately following initial contact was used to calculate the area under the RMS curve using the same normalization procedures and number of strides as previously described.

#### 2.5.4. Onset timing of sEMG relative to initial contact

In order to determine the onset of muscle activation relative to initial contact, an activation threshold was set at 10 standard deviations above the quiet standing standard deviation (Feger et al., 2014). This was then applied to the real-time data through an algorithm, simplifying the data to values above threshold as "1" and values below the threshold as "0". The amount of time between initial contact and the earliest indication of muscle activation (either prior to or after initial contact) was measured for each muscle. The mean across 9 strides for each muscle was used for statistical analysis.

#### 2.5.5. Percent activation time

Using the same algorithm to determine activation as described previously, a percentage of activation over each stride cycle was calculated. The mean across strides was used for statistical analysis.

### 2.6. Statistical analysis

Paired *t*-tests were used to analyze differences in sEMG measures of pre- and post-contact amplitude, onset timing and percent activation time for the AT, PL, LG, RF, BF, GM between conditions with an alpha level set *a priori* at $p < 0.05$. Per contemporary statistical recommendations, we elected to not perform any correction for multiple comparisons (Hopkins, Marshall, Batterham, & Hanin, 2009). Additionally, to evaluate the magnitude of difference between the brace and no brace conditions, Cohen's *d* effect sizes and 95% confidence intervals were also calculated. Effect sizes were interpreted as ≥0.80 was large, ≥0.50 was moderate, ≥0.20 was small and <0.20 was minimal.

## 3. Results

### 3.1. Amplitude measures

The descriptive measures for the sEMG amplitude results are shown in Tables 2 and 3. The pre-contact amplitude of the PL was significantly lower in the brace condition when compared to the no brace condition ($0.36 \pm 0.18$ vs. $0.46 \pm 0.23$, $p = 0.02$) Table 2. There

**Table 2**
Results of 100 ms pre-initial contact sEMG amplitudes of no brace and brace during walking.

| Muscle | Pre-initial contact mean (standard deviation) | | Paired *T*-test | Pre-initial contact effect size (lower limit, upper limit) |
|---|---|---|---|---|
| | No brace | Brace | *p*-Value | No brace–brace |
| Anterior tibialis | 2.66 (2.33) | 2.43 (2.07) | 0.19 | −0.10 (−0.81, 0.62) |
| Peroneus longus | 0.46 (0.23) | 0.36 (0.18) | 0.02[a] | −0.43 (−1.15, 0.30) |
| Lateral gastrocnemius | 0.21 (0.11) | 0.17 (0.14) | 0.18 | −0.35 (−1.07, 0.37) |
| Rectus femoris | 0.75 (0.44) | 0.65 (0.42) | 0.06 | −0.23 (−0.94, 0.49) |
| Biceps femoris | 1.42 (1.09) | 1.43 (1.15) | 0.78 | −0.02 (−0.70, 0.73) |
| Gluteus medius | 0.54 (0.49) | 0.41 (0.31) | 0.11 | −0.26 (−0.98, 0.46) |

Negative effect size represents higher amplitude in the no brace condition when compared to the brace condition.
[a] Brace condition had significantly higher amplitude when compared to the no brace condition.

**Table 3**
Results of 200 ms post-initial contact sEMG amplitudes of no brace and brace during walking.

| Muscle | Post-initial contact mean (standard deviation) | | Paired *T*-test | Post-initial contact effect size (lower limit, upper limit) |
|---|---|---|---|---|
| | No brace | Brace | *p*-Value | No brace–brace |
| Anterior tibialis | 3.59 (3.27) | 3.23 (2.80) | 0.19 | −0.11 (−0.82, 0.61) |
| Peroneus longus | 1.22 (0.64) | 1.00 (0.54) | 0.09 | −0.35 (−1.07, 0.37) |
| Lateral gastrocnemius | 0.56 (0.35) | 0.62 (0.70) | 0.74 | 0.18 (−0.53, 0.90) |
| Rectus femoris | 2.17 (1.53) | 1.95 (1.27) | 0.15 | −0.14 (−0.86, 0.57) |
| Biceps femoris | 1.21 (0.96) | 1.25 (0.98) | 0.67 | 0.03 (−0.68, 0.75) |
| Gluteus medius | 1.93 (1.90) | 1.85 (2.03) | 0.45 | −0.04 (−0.76,0.68) |

Negative effect size represents higher amplitude in the no brace condition when compared to the brace condition.

were no significant differences for any of the muscles between conditions in the post-contact amplitude measures Table 3.

### 3.2. Onset timing

The descriptive measures for the sEMG onset timing are shown in Table 4. The AT, PL, RF and GM were activated significantly later relative to contact in the brace condition when compared to the no brace condition (AT: $-0.32 \pm 0.02$ vs. $-0.36 \pm 0.16$ sec, $p = 0.02$; PL: $0.02 \pm 0.12$ vs. $-0.07 \pm 0.10$ sec, $p = 0.02$; RF: $-0.07 \pm 0.06$ vs. $-0.14 \pm 0.15$ sec, $p = 0.03$; GM: $050 \pm 0.082$ vs. $-0.08 \pm 0.10$ sec, $p = 0.01$).

### 3.3. Percent activation time

The descriptive measures for the sEMG percent activation time results are shown in Table 5. The PL and RF were activated for a significantly shorter percentage across the stride cycle in the brace condition when compared to the no brace condition (PL: $28.67 \pm 9.35$ vs. $36.04 \pm 10.26\%$, $p = 0.002$; RF: $30.79 \pm 18.86$ vs. $36.90 \pm 24.52\%$, $p = 0.05$).

## 4. Discussion

Our results demonstrate that lace-up ankle braces cause alterations in lower extremity muscle activation during walking in physically active young adults with CAI. Specifically, bracing caused a decrease in the PL pre-contact amplitude, delayed activation in the AT, PL, RF and GM, and overall decreased activation of the PL and RF throughout the stride cycle. Our results provide insight into the potential physiologic mechanisms by which lace-up ankle braces may reduce the risk of ankle sprains.

Ankle bracing has shown the ability to prevent initial and recurrent ankle sprains (McGuine et al., 2011, 2012). Whether it is a result of neuromuscular alterations, gait alterations or mechanical support is uncertain. One injury prevention theory that has been suggested is the idea of a "feed-forward" mechanism (Forestier & Toschi, 2005). This theory suggests that preemptive firing of the PL may be beneficial in stiffening the ankle joint and providing increased dynamic control to make up for lacking static stability (i.e. ligamentous support) (Forestier & Toschi, 2005). Ankle bracing has been thought to contribute to this mechanism through increased stimulation of cutaneous mechanoreceptors (Cordova & Ingersoll, 2003).

The feed-forward theory however, has recently been challenged (Feger et al., 2014). Feger et al. (2014) found individuals with CAI to have earlier activation of the PL and RF, as well increased overall activation of the PL throughout the stride cycle, indicating that CAI patients have neuromuscular adaptations prior to initial contact. The function of the PL in healthy people is to pull the first ray toward the ground during the loading response and pronate the foot which occurs after approximately 5–10% of the stride cycle has passed (Feger et al., 2014; Forestier & Toschi, 2005). Feger et al. (2014) postulated that earlier, prolonged activation of the PL may have negative consequences for individuals with CAI possibly leading to a decreased ability to respond to perturbation, fatigue and chronic instability episodes. However, Forestier and Toschi (2005) believe early activity of the peroneus longus during gait may be beneficial in providing more dynamic stability to the ankle joint, which may protect against lateral ankle sprains. Based on both of these theories, we believe if a patient does not use an external supportive device, clinicians should try to exploit the early activation of the peroneus longus to maximize dynamic stability.

The importance of the PL in providing dynamic stability to the lateral ankle complex is undeniable (Ashton-Miller, Ottaviani, Hutchinson, & Wojtys, 1996). On average, individuals with CAI activate their PL 70 ms prior to initial contact compared to healthy individuals which activate 160 ms after initial contact (Feger et al., 2014). Our study found that bracing shifted PL activation to 20 ms prior to initial contact. This finding is not only statistically significant but also clinically meaningful with a large effect size ($d = 0.84 \pm 0.74$). Furthermore, the PL was activated for 36% of the stride cycle in the CAI group compared to controls at 23%. Our study found PL activation was reduced to 29% of the stride cycle with the

**Table 4**
Results of onset timing in seconds relative to initial contact of no brace and brace during walking.

| Muscle | Onset time mean (standard deviation) | Onset time mean (standard deviation) | Paired *T*-test | Effect size (lower limit, upper limit) |
|---|---|---|---|---|
| | No brace | Brace | *p*-Value | No brace–brace |
| Anterior tibialis | −0.36 (0.16) | −0.32 (0.15) | 0.02[a] | 0.24 (−0.48, 0.96) |
| Peroneus longus | −0.07 (0.10) | −0.02 (0.12) | 0.02[a] | 0.84 (0.09, 1.58) |
| Lateral gastrocnemius | 0.18 (0.14) | 0.20 (0.15) | 0.62 | 0.18 (−0.54, 0.89) |
| Rectus femoris | −0.14 (0.15) | −0.07 (0.06) | 0.03[a] | 0.49 (−0.23, 1.22) |
| Biceps femoris | −0.23 (0.10) | −0.19 (0.06) | 0.15 | 0.33 (−0.39, 1.05) |
| Gluteus medius | −0.08 (0.10) | −0.05 (0.08) | 0.01[a] | 0.30 (−0.42, 1.02) |

Negative values indicate pre-activation relative to initial contact.
Positive effect size represents earlier onset time in the no brace condition when compared to the brace condition.
[a] Brace condition had significantly later onset time when compared to the no brace condition.

**Table 5**
Results of percent of activation time across the stride cycle of no brace and brace during walking.

| Muscle | Onset time mean % (standard deviation) | Onset time mean % (standard deviation) | Paired *T*-test | Effect size (lower limit, upper limit) |
|---|---|---|---|---|
| | No brace | Brace | *p*-Value | No brace−brace |
| Anterior tibialis | 53.14 (26.71) | 51.16 (25.63) | 0.40 | −0.07 (−0.79, 0.64) |
| Peroneus longus | 36.04 (10.26) | 28.67 (9.35) | 0.002[a] | −0.72 (−1.46, 0.02) |
| Lateral gastrocnemius | 21.07 (12.19) | 21.54 (14.85) | 0.92 | 0.04 (−0.68, 0.75) |
| Rectus femoris | 36.90 (24.52) | 30.79 (18.86) | 0.05[a] | −0.25 (−0.97, 0.47) |
| Biceps femoris | 29.39 (19.60) | 25.13 (15.91) | 0.28 | 0.22 (−0.93, 0.50) |
| Gluteus medius | 27.18 (25.23) | 23.69 (25.01) | 0.31 | −0.14 (−0.85, 0.58) |

Negative effect size represents higher percent activation across the stride cycle in the no brace condition when compared to the brace condition.
[a] Brace condition had significantly lower percent activation across the stride cycle when compared to the no brace condition.

brace, again, an apparent shift toward the norm with a moderate effect size ($d = -0.72 \pm 0.74$). Our study also correlated with Feger et al. (2014) findings with CAI individuals activating their RF earlier than controls, with delayed activation of the RF while braced. In every area where a difference was found between healthy and CAI groups (Feger et al., 2014), our results demonstrated alterations with the brace application shifting CAI group toward the norm seen in healthy controls.

To understand how bracing influences muscle activation, it may be important to think about what is occurring at the ankle joint as a result of the brace, prior to contact with the ground (DiStefano et al., 2008; Eils & Rosenbaum, 2003). Gait analysis studies have shown that individuals with CAI have a more inverted foot position prior to and immediately after initial contact (Monaghan, Delahunt, & Caulfield, 2006). Conversely, braced individuals have been found to have a more everted foot position prior to contact (Nishikawa, Kurosaka, et al., 2002). This repositioning of the foot also appears to carry through to loading and stance phases of gait showing increases in maximum pronation and pronation velocity (Nishikawa, Kurosaka, et al., 2002). It may simply be the detection of this repositioning by ankle joint mechanoreceptors that causes the individual to alter their muscle activation pattern.

The muscle activation changes due to bracing could also be related to increased afferent signal to the central nervous system (CNS) via cutaneous mechanoreceptors, which may also have an indirect effect on joint mechanoreceptors (Cordova & Ingersoll, 2003). This increased afferent signal may be why ankle bracing has shown to effect reflex amplitude (Cordova & Ingersoll, 2003; M. L. Cordova et al., 2000) and timing (Shima et al., 2005) as well as motoneuron pool excitability (Nishikawa & Grabiner, 1999). Feuerbach et al. (1994) performed a study evaluating ankle joint proprioception before and after ankle ligament anesthesia and found that bracing improved joint position sense under both conditions. While Feuerbach et al. (1994) attempted to recreate the articular de-afferentation environment similar to that which was proposed by Freeman (Freeman et al., 1965) as a source of functional ankle instability, this experimental model was performed on otherwise healthy participants. Therefore, all sequelae that occur as a result of lateral ankle sprain could not be replicated. If improvements in joint position sense after bracing are seen in the healthy population, it is reasonable to presume that similar changes may be seen in the pathological population.

Ankle braces, as a form of mechanical support to the ankle complex, have shown an ability to decrease range of motion in plantarflexion, inversion, and dorsiflexion (M. Cordova, Ingersoll, & LeBlanc, 2000; DiStefano et al., 2008) and to slow the rate of inversion (Nishikawa, Kurosaka, et al., 2000; Nishikawa, Ozaki, Mizuno, & Grabiner, 2002) in response to a perturbation. The range of motion measurements used in these studies, however, likely do not provide an accurate depiction of what occurs in a real-time ankle injury. Further research simultaneously assessing the mechanical and sensorimotor effects of ankle braces may provide further insights into how bracing reduces the incidence of ankle injuries.

Ankle bracing has not been shown to have detrimental effects on knee or hip kinematics or lower extremity ground reaction forces (DiStefano et al., 2008) and has not been shown to increase injury risk at the knee or hip (McGuine et al., 2011, 2012). Furthermore, prolonged use of ankle bracing has not been shown to have an effect on kinematics (DiStefano et al., 2008), ground reaction forces (DiStefano et al., 2008), or PL function (M. L. Cordova et al., 2000) in healthy individuals. It has yet to be determined if there are any effects, positive or negative, on prolonged brace use in those with CAI.

Overall, ankle bracing is a cost and time effective (Soboroff, Pappius, & Komaroff, 1984) way to possibly modify muscle activation patterns to what they were prior to initial ankle sprain. This may improve the efficiency of both the dynamic ankle stabilizers and CNS. Our study was limited to a walking task and the translatability of these results to more challenging tasks such as running are unknown. In addition, our results can only be generalized to CAI patients as we operationally defined the pathology. We did not differentiate between mechanical instability and functional instability as a result of the participant's lateral ankle sprains. Therefore, we do not know if muscle activity responds differently to an ankle brace for individuals who have or do not have an unstable joint as a result of mechanical damage to the ligaments. Future research is warranted to evaluate the effect of bracing on sEMG activity at higher gait speeds and during other functional tasks such as cutting and jumping. It would also be beneficial to investigate and correlate data between kinetic variables, kinematic variables and muscle activity.

## 5. Conclusions

We found that in active individuals with CAI, a lace-up ankle brace reduced PL pre-contact amplitude, delayed activation of the AT, PL, RF and GM and decreased total on-time of the PL and RF throughout the stride cycle during treadmill walking. These differences between conditions may suggest that the alteration in muscle function may contribute to the mechanism in which lace-up braces decrease the recurrence of ankle sprains in people with CAI. Clinicians should be aware of these changes when prescribing braces, as it may relate to the mechanism in which braces decrease sprains.

*Conflict of interest*
None declared.

*Ethical approval*
The University's Institution Review Board for Health Sciences (IRB-HSR-15877) has approved this project.

*Funding*
None declared.

## Acknowledgment

We would like to acknowledge Mark Feger for his contributions to data collection and analysis.

## References

Ashton-Miller, J. A., Ottaviani, R. A., Hutchinson, C., & Wojtys, E. M. (1996). What best protects the inverted weightbearing ankle against further inversion? Evertor muscle strength compares favorably with shoe height, athletic tape, and three orthoses. *The American Journal of Sports Medicine, 24*(6), 800–809.

Carcia, C. R., Martin, R. R. L., & Drouin, J. M. (2008). Validity of the foot and ankle ability measure in athletes with chronic ankle instability. *Journal of Athletic Training, 43*(2), 179.

Chinn, L., Dicharry, J., & Hertel, J. (2013). Ankle kinematics of individuals with chronic ankle instability while walking and jogging on a treadmill in shoes. *Physical Therapy in Sport*.

Cordova, M. L., Armstrong, C. W., Rankin, J. M., & Yeasting, R. A. (1998). Ground reaction forces and EMG activity with ankle bracing during inversion stress. *Medicine and Science in Sports and Exercise, 30*(9), 1363–1370.

Cordova, M. L., Cardona, C. V., Ingersoll, C. D., & Sandrey, M. A. (2000). Long-term ankle brace use does not affect peroneus longus muscle latency during sudden inversion in normal subjects. *Journal of Athletic Training, 35*(4), 407.

Cordova, M., & Ingersoll, C. (2003). Peroneus longus stretch reflex amplitude increases after ankle brace application. *British Journal of Sports Medicine, 37*(3), 258–262.

Cordova, M., Ingersoll, C., & LeBlanc, M. (2000). Influence of ankle support on joint range of motion before and after exercise: a meta-analysis. *The Journal of Orthopaedic and Sports Physical Therapy, 30*(4), 170–177. discussion 178–182.

Delahunt, E., Coughlan, G. F., Caulfield, B., Nightingale, E. J., Lin, C., & Hiller, C. E. (2010). Inclusion criteria when investigating insufficiencies in chronic ankle instability. *Medicine and Science in Sports and Exercise, 42*(11), 2106.

Delahunt, E., Monaghan, K., & Caulfield, B. (2006a). Altered neuromuscular control and ankle joint kinematics during walking in subjects with functional instability of the ankle joint. *The American Journal of Sports Medicine, 34*(12), 1970.

Delahunt, E., Monaghan, K., & Caulfield, B. (2006b). Changes in lower limb kinematics, kinetics, and muscle activity in subjects with functional instability of the ankle joint during a single leg drop jump. *Journal of Orthopaedic Research, 24*(10), 1991–2000.

DiStefano, L. J., Padua, D. A., Brown, C. N., & Guskiewicz, K. M. (2008). Lower extremity kinematics and ground reaction forces after prophylactic lace-up ankle bracing. *Journal of Athletic Training, 43*(3), 234.

Docherty, C. L., Valovich McLeod, T. C., & Shultz, S. J. (2006). Postural control deficits in participants with functional ankle instability as measured by the balance error scoring system. *Clinical Journal of Sport Medicine, 16*(3), 203.

Eils, E., & Rosenbaum, D. (2003). The main function of ankle braces is to control the joint position before landing. *Foot & Ankle International, 24*(3), 263–268.

Evans, T., Hertel, J., & Sebastianelli, W. (2004). Bilateral deficits in postural control following lateral ankle sprain. *Foot & Ankle International, 25*(11), 833–839.

Feger, M., Donovan, L., Hart, J., & Hertel, J. (2014). Lower extremity muscle activation in patients with and without chronic ankle instability. *Journal of Athletic Training*.

Feuerbach, J. W., Grabiner, M. D., Koh, T. J., & Weiker, G. G. (1994). Effect of an ankle orthosis and ankle ligament anesthesia on ankle joint proprioception. *The American Journal of Sports Medicine, 22*(2), 223–229.

Forestier, N., & Toschi, P. (2005). The effects of an ankle destabilization device on muscular activity while walking. *International Journal of Sports Medicine, 26*(6), 464.

Freeman, M., Dean, M., & Hanham, I. (1965). The etiology and prevention of functional instability of the foot. *Journal of Bone and Joint Surgery-British Volume, 47*(4), 678.

Hertel, J. (2002). Functional anatomy, pathomechanics, and pathophysiology of lateral ankle instability. *Journal of Athletic Training, 37*(4), 364.

Hertel, J. (2008). Sensorimotor deficits with ankle sprains and chronic ankle instability. *Clinics in Sports Medicine, 27*(3), 353–370.

Hiller, C. E., Kilbreath, S. L., & Refshauge, K. M. (2011). Chronic ankle instability: evolution of the model. *Journal of Athletic Training, 46*(2), 133–141.

Hootman, J., Dick, R., & Agel, J. (2007). Epidemiology of collegiate injuries for 15 sports: summary and recommendations for injury prevention initiatives. *Journal of Athletic Training, 42*(2), 311.

Hopkins, W., Marshall, S., Batterham, A., & Hanin, J. (2009). Progressive statistics for studies in sports medicine and exercise science. *Medicine+ Science in Sports+ Exercise, 41*(1), 3.

Konrad, P. (2005). *The ABC of EMG: A practical introduction to kinesiological electromyography*. Version 1.0. USA: Noraxon Inc..

Lehman, G. J., & McGill, S. M. (1999). The importance of normalization in the interpretation of surface electromyography: a proof of principle. *Journal of Manipulative and Physiological Therapeutics, 22*(7), 444–446.

McGuine, T. A., Brooks, A., & Hetzel, S. (2011). The effect of lace-up ankle braces on injury rates in high school basketball players. *The American Journal of Sports Medicine, 39*(9), 1840–1848.

McGuine, T. A., Hetzel, S., Wilson, J., & Brooks, A. (2012). The effect of lace-up ankle braces on injury rates in high school football players. *The American Journal of Sports Medicine, 40*(1), 49–57.

McKay, G., Goldie, P., Payne, W., & Oakes, B. (2001). Ankle injuries in basketball: injury rate and risk factors. *British Journal of Sports Medicine, 35*(2), 103.

Monaghan, K., Delahunt, E., & Caulfield, B. (2006). Ankle function during gait in patients with chronic ankle instability compared to controls. *Clinical Biomechanics, 21*(2), 168–174.

Munn, J., Sullivan, S. J., & Schneiders, A. G. (2010). Evidence of sensorimotor deficits in functional ankle instability: a systematic review with meta-analysis. *Journal of Science and Medicine in Sport, 13*(1), 2–12.

Nishikawa, T., & Grabiner, M. (1999). Peroneal motoneuron excitability increases immediately following application of a semirigid ankle brace. *The Journal of Orthopaedic and Sports Physical Therapy, 29*(3), 168.

Nishikawa, T., Kurosaka, K., Mizuno, K., & Grabiner, M. (2000). Protection and performance effects of ankle bracing. *International Orthopaedics, 24*(5), 285–288.

Nishikawa, T., Kurosaka, M., Yoshiya, S., Lundin, T., & Grabiner, M. (2002). Effects of prophylactic ankle supports on pronation during gait. *International Orthopaedics, 26*(6), 381–385.

Nishikawa, T., Ozaki, T., Mizuno, K., & Grabiner, M. D. (2002). Increased reflex activation of the peroneus longus following application of an ankle brace declines over time. *Journal of Orthopaedic Research, 20*(6), 1323–1326.

van Rijn, R. M., van Os, A. G., Bernsen, R., Luijsterburg, P. A., Koes, B. W., & Bierma-Zeinstra, S. (2008). What is the clinical course of acute ankle sprains? A systematic literature review. *The American Journal of Medicine, 121*(4), 324–331.

Sefton, J., Hicks-Little, C., Koceja, D., & Cordova, M. (2007). Effect of inversion and ankle bracing on peroneus longus Hoffmann reflex. *Scandinavian Journal of Medicine & Science in Sports, 17*(5), 539–546.

Shima, N., Maeda, A., & Hirohashi, K. (2005). Delayed latency of peroneal reflex to sudden inversion with ankle taping or bracing. *International Journal of Sports Medicine, 26*(06), 476–480.

Soboroff, S. H., Pappius, E. M., & Komaroff, A. L. (1984). Benefits, risks, and costs of alternative approaches to the evaluation and treatment of severe ankle sprain. *Clinical Orthopaedics and Related Research, 183*, 160.

Waterman, B. R., Owens, B. D., Davey, S., Zacchilli, M. A., & Belmont, P. J. (2010). The epidemiology of ankle sprains in the United States. *Journal of Bone and Joint Surgery, American Volume, 92*(13), 2279–2284.